

Water Quality Index (WQI) Prediction Pipeline

Final Documentation

1. Introduction

This document outlines the final methodology and machine learning pipeline developed to predict the Water Quality Index (WQI) Suitability Class for the Sibalom River. By leveraging 6 years of historical data from the Jalaur River System (JRS-WQMA), we utilized Transfer Learning and Synthetic Minority Over-sampling Technique (SMOTE) to build a highly accurate machine learning model capable of predicting water quality based on six critical parameters.

2. Statement of the Problem

The Sibalom River currently lacks continuous water quality monitoring. The core problem is how to accurately and efficiently determine the Water Quality Index (WQI) Suitability Class of the river using machine learning. Initially, the goal was to bypass laboratory testing by only using field-deployable meters (pH, DO, Temperature). However, mathematical analysis proved that bacterial contamination (Fecal Coliform) is the primary driver of 'High Risk' classifications in the region, and cannot be predicted accurately using only pH and Temperature. Therefore, the problem shifted to building an optimized ML model that utilizes all 6 parameters (including lab tests) to accurately classify water quality despite the severe class imbalance (97% of local rivers being High Risk).

3. Pipeline Flowchart & Methodology

The end-to-end pipeline operates in the following sequence:

- Step 1:** Data Collection (Raw PDF extraction to CSV)
- Step 2:** Preprocessing (KNN Imputation & Isolation Forest Outlier Removal)
- Step 3:** Feature Engineering (Calculation of DO/Temp Ratio and pH Deviation)
- Step 4:** WQI Computation (Using DENR DAO 2016-08 formula)
- Step 5:** Classification (Excellent, Good, Fair, Poor, High Risk)
- Step 6:** Dataset Balancing (SMOTE to generate minority class synthetic data)
- Step 7:** ML Model Training (Grid Search CV for optimal hyperparameters)
- Step 8:** Deployment & Simulation (Predicting Sibalom River quality)

4. Machine Learning Models Comparative Table

Algorithm	Accuracy	F1-Score	Description
Random Forest	98.01%	97.63%	Baseline ensemble. Handles non-linear relationships well but struggles with extreme minority classes without SMOTE.
XGBoost	98.01%	98.07%	Gradient boosting algorithm. Highly accurate and resistant to overfitting via regularization.

MLP (Neural Net)	96.69%	97.56%	Multi-layer perceptron. Chosen deployment model. Natively Maps the multi-dimensional dataset perfectly.
Support Vector Machine	83.44%	88.14%	Finds optimal hyperplanes. Underperformed due to heavily overlapping feature boundaries in water data.

5. Machine Learning Algorithm to Use: XGBoost

The XGBoost algorithm achieved an accuracy of 98.01% and was capable of effectively mapping the complex mathematical relationships in the dataset. Because the training dataset was heavily imbalanced (86% High Risk), the ML model demonstrates how machine learning adapts to heavily skewed environmental data. The resulting F1-Score of 96.59% guarantees extremely high reliability across the WQI spectrum, making it the optimal choice for analyzing the complex chemical interactions of the Sibalom River samples.

6. Evaluation Output

The XGBoost model was evaluated on a SMOTE-balanced training set (80%), and then tested against the true distribution of the remaining 20% holdout set. The model correctly classified 100% of 'High Risk' samples and 100% of 'Excellent' samples, proving its robustness. It demonstrated strong predictive capabilities despite the severe 86% class imbalance of the original Jalaur river dataset toward High Risk. The resulting F1-Score of 96.59% guarantees high reliability for field deployment and environmental analysis.

7. Flow of the Testing Process

The physical testing and deployment process follows these steps:

1. Sample Collection: Water samples are collected from the Sibalom River.
2. Field Testing: Portable meters are used on-site to measure pH, DO, and Temperature.
3. Lab Testing: The sample is sent to a laboratory to measure BOD, TSS, and Fecal Coliform.
4. Data Entry: The user opens the Web Dashboard and inputs the 6 parameters. Missing data can be marked as 'N/A'.
5. KNN Imputation: If 'N/A' is entered, the backend intelligently estimates the missing value using K-Nearest Neighbors.
6. ML Inference: The XGBoost model processes the features and outputs an instant WQI Suitability Classification.

8. Evaluation and Comparative Analysis Methodology

To scientifically validate the machine learning model on the new target domain (Sibalom River), the following comparative analysis framework will be executed:

Step 1: Data Gathering (Field & Lab)

Water samples are physically collected from various sites along the Sibalom River. Portable meters immediately measure pH, DO, and Temperature, while the remaining samples are sent to an accredited laboratory to test for BOD, TSS, and Fecal Coliform.

Step 2: Ground Truth Computation (The Math)

The 6 laboratory parameters for each sample are manually calculated using the exact DENR DAO 2016-08 Weighted Arithmetic Water Quality Index formula. The resulting numerical score dictates the absolute 'Ground Truth' classification (e.g., a math score of 62.4 = Fair).

Step 3: AI Inference (The Prediction)

The exact same 6 parameters are fed into the XGBoost Machine Learning model via the web

application. The model processes the chemical signatures and outputs its predicted WQI classification based solely on what it learned from the historical Jalaur dataset.

Step 4: Comparative Matrix

The Ground Truth classes are placed side-by-side against the AI Predictions in a Confusion Matrix to track exact matches and deviations.

Step 5: Final Analysis

Researchers will analyze the accuracy percentage of the model. More importantly, any deviations will be analyzed in the context of dataset bias (e.g., investigating if the AI's exposure to the heavily polluted 86% High Risk Jalaur dataset caused it to misclassify moderately clean Sibalom samples).

9. Glossary of Machine Learning Concepts

Transfer Learning: A machine learning technique where a model trained on one task or dataset (e.g., Jalaur River) is repurposed or applied to a different but related task (e.g., Sibalom River). This is crucial when the target river lacks historical training data.

Preprocessing: The fundamental step of cleaning and organizing raw data before feeding it into a machine learning model. This includes handling missing values, removing anomalies, and scaling numbers so the algorithms can process them efficiently.

KNN Imputation: K-Nearest Neighbors (KNN) Imputation is an algorithm used to fill in missing data points (like a missing pH reading). It looks at the 'k' most similar data rows (neighbors) and averages their values to estimate and replace the missing number.

Isolation Forest Outlier Removal: An anomaly detection algorithm that builds random decision trees to isolate individual data points. Normal data points require many splits to be isolated, whereas extreme outliers are isolated very quickly. We use this to remove faulty sensor readings or mathematically impossible data spikes.

Dataset Balancing (SMOTE): Synthetic Minority Over-sampling Technique (SMOTE) is used to balance uneven datasets. Since 97% of our river data was 'High Risk', the model would normally just guess 'High Risk' every time. SMOTE mathematically generates fake (but highly realistic) synthetic data points for the rare classes (like 'Good' or 'Poor') so the ML model can learn exactly what clean water looks like.

Classification: A type of machine learning task where the goal is to predict a discrete category or label for a given input. In this study, the model classifies water into categories like 'Excellent', 'Fair', or 'High Risk'.

Training & Validation: After rigorous testing involving 5-Fold Stratified GridSearchCV (optimizing 180 hyperparameter combinations for the Neural Network and 90 combinations for XGBoost), the XGBoost model proved to be the most robust and accurate for our deployment space. While the highly-tuned Neural Network achieved a commendable 95.70% F1-Score, the XGBoost algorithm captured the intricate non-linear dynamics of the synthetic clean baseline metrics more effectively, resulting in a near-perfect 98.07% F1-Score on the out-of-distribution test set. Therefore, the final prediction pipeline deployed to the Streamlit/Next.js backend strictly utilizes the optimized XGBClassifier to perform inference on field-deployed hardware readings.

XGBoost: Extreme Gradient Boosting is an advanced algorithm that also builds decision trees, but it builds them sequentially. Each new tree specifically tries to correct the errors made by the previous trees, resulting in a highly optimized and powerful predictive model.

MLP (Multilayer Perceptron): A type of Artificial Neural Network inspired by the human brain. It consists of multiple layers of interconnected 'neurons' that pass data through mathematical activation functions. It is excellent at finding deeply hidden, non-linear patterns in complex data.

SVM (Support Vector Machine): An algorithm that attempts to find the optimal mathematical 'line' or 'hyperplane' in multi-dimensional space that perfectly separates different classes of data. While powerful, it struggled to separate the highly imbalanced classes in our specific water quality dataset.

10. Student Research Guide: Core Concepts to Master

To fully understand and defend this project, students should dive deeper into the following core concepts. Mastering these topics provides a complete grasp of both the environmental science and the advanced machine learning techniques used.

Transfer Learning & Domain Adaptation

What to research: How machine learning models learn patterns in one domain (Jalaur River) and apply them to a different, related domain (Sibalom River).

Why it matters: This is the core justification of the entire study. You need to explain why it is scientifically valid to train a model on one river and test it on another.

Dealing with Imbalanced Data: The SMOTE Algorithm

What to research: Synthetic Minority Over-sampling Technique (SMOTE). How does it use mathematics (K-Nearest Neighbors) to generate fake but realistic data points?

Why it matters: 97% of your original data was 'High Risk'. Without SMOTE, the model would have failed. Understanding how SMOTE saved the dataset is critical for your methodology defense.

Machine Learning Algorithms: MLP & Ensemble Learning

What to research: What is a Neural Network (MLP)? What is Ensemble Learning? How does XGBoost build sequential trees that learn from each other's mistakes?

Why it matters: MLP was the deployment model (96.69% accuracy) because Neural Networks map complex multidimensional decision spaces natively.

Ground Truth & Model Bias

What to research: How does an AI model react when trained on data that is mostly one class (86% High Risk)?

Why it matters: The highlight of the study is exploring how the ML model learned to identify pollution based on a skewed dataset, and analyzing how it predicts borderline classes compared to standard mathematical formulas.

Data Imputation: K-Nearest Neighbors (KNN)

What to research: How does the KNN algorithm calculate Euclidean distance to find the most similar data points?

Why it matters: When the DENR PDF data had blank/missing values for pH or DO, you didn't just delete the row or use the average. You used KNN to mathematically estimate what the missing value should be based on its closest neighbors.

The Mathematics of Feature Engineering

What to research: Pearson Correlation and thermodynamic relationships in water (e.g., why does Dissolved Oxygen capacity drop when Temperature rises?).

Why it matters: You explicitly engineered two new features: DO_Temp_Ratio and pH_Deviation. You need to explain why providing these explicit mathematical relationships helped the ML model learn faster.

Water Quality Index (WQI) Formulas

What to research: The Weighted Arithmetic Water Quality Index method and the DENR DAO 2016-08 standards.

Why it matters: You need to understand how the raw lab numbers (like 140 MPN Coliform) were converted into the WQI Score that determined the final Suitability Class (Excellent to High Risk).

Evaluation Metrics Beyond Accuracy

What to research: Why is plain Accuracy dangerous when you have an imbalanced dataset? What are Precision, Recall, and the F1-Score?

Why it matters: If a model simply guesses 'High Risk' every time on a dataset that is 97% High Risk, it will get 97% Accuracy but is utterly useless. You must explain why the F1-Score is the true measure of your model's success.

11. Mathematical & Computational Chemistry Framework

As a study deeply rooted in computational chemistry and data science, several explicit mathematical and statistical computations were applied to transform raw chemical concentrations into a predictive machine learning framework. The following mathematical computations were utilized:

1. Quality Rating Scale (q_i)

Each chemical parameter was mathematically normalized into a standard rating scale using the formula:

$$q_i = 100 \times \frac{V_i - V_{ideal}}{S_i - V_{ideal}}$$

This transforms raw concentrations (V_i) against chemical standards (S_i) and ideal values (V_{ideal}) into a unitless metric.

2. Weighted Arithmetic Water Quality Index (WQI)

The aggregation of multiple chemical quality ratings into a single, definitive index score. The formula used was:

$$WQI = \frac{\sum (q_i \times w_i)}{\sum w_i}$$

where ' w_i ' represents the mathematical weight or importance of each chemical parameter.

3. Information Entropy Weighting

Rather than subjectively assigning weights, Information Theory (Entropy) mathematics was used to calculate objective weights for each chemical parameter. This statistical computation measures the variance and dispersion of a chemical's concentration across the dataset to determine its true informative value.

4. Pearson Correlation Coefficient

A mathematical computation used to measure the linear correlation between individual chemical variables and the final WQI. This statistical analysis helped refine the final Combined Weights used in the feature engineering process.

5. Euclidean Distance (KNN Imputation)

To handle missing data (e.g., a missing DO reading), the K-Nearest Neighbors algorithm utilized the Euclidean Distance formula in multi-dimensional chemical space to find the most chemically similar river samples and average their values:

$$d(p, q) = \sqrt{\sum (p_i - q_i)^2}$$

6. Thermodynamic Ratios (DO/Temp Ratio)

An engineered computation that explicitly models the inverse non-linear relationship between water temperature and the physical solubility of oxygen. By dividing Dissolved Oxygen by Temperature, we fed the ML model a direct thermodynamic constraint.

7. pH Deviation

A simple but critical mathematical absolute distance function:

$$\Delta pH = |pH - 7.0|$$

Since pH is logarithmic, modeling the absolute deviation from neutral proved more mathematically useful for the machine learning algorithms than raw pH values alone.